

WQD7013 – Statistics for Data Science
ASSIGNMENT 1
Group 7

Name	Student ID
Pan Hui Xin	25090880
Ng Jing Lee	24215833
Nur Aliah Natasya	25085928
Anisah Maisarah Binti Ahmad Nasaruddin	S2121975
Nurin Miza Afiqah Binti Andrie Dazlee	s2154141
Mohd Khairi bin Yusof	S2191607

Task A – Dataset Audit and Variable Classification

A1. Structural Inspection

A1.1 Load Dataset

The csv file loads as expected with a dataset containing 30450 rows $\times$ 12 columns that satisfy the minimum size requirement for our analysis and matching the Kaggle dataset description.

A1.2 Data Types

This section is to investigate the loaded dataset, where categorical variables such as `country\_name`, `city\_name` and `Date` are stored as `object` types. Numerical variables such as pollutant concentrations (e.g., `PM2.5`, `PM10`, `NO2`, `SO2`, `CO`, `O3`, `Temperature`, `Humidity`, `Wind Speed`) are stored as numeric types `float64`. Overall, most numerical variables are correctly identified. However, some categorical variables such as `City`, `Country`, and temporal variables for `Date` need further attention.

A1.3 Missing Value

From the A1.3 output, the missing values were assessed for each column. It is observed that the air pollutant variables including `PM2.5`, `PM10`, `NO2`, `SO2`, `CO`, and `O3` contain missing values. Specifically:

- `PM2.5` has 1552 missing values (5.10%)
- `PM10` has 1561 missing values (5.13%)
- `NO2` has 1593 missing values (5.23%)
- `SO2` has 1475 missing values (4.84%)
- `CO` has 1496 missing values (4.91%)
- `O3` has 1600 missing values (5.26%)

In contrast, variables such as `City`, `Country`, `Date`, `Temperature`, `Humidity`, and `Wind Speed` have no missing values (0%). In short, the dataset can be considered partially incomplete, with missing values concentrated in pollutant measures. The proportion of missing data is relatively moderate (about 5%) and the proper handling strategies such as imputation or deletion may be required in future analyses.

A1.4 Duplicate Rows

From the A1.4 output, a total of 10495 duplicate rows were identified in the dataset.

A1.5 Column with Wrong Inferred dtype

From the A1.5 output, the following columns `City`, `Country`, and `Date` were identified as having incorrect inferred data types and require conversion to ensure accurate and efficient analysis.

Column	(a) Python Inferred Type	(b) Correct Type	(c) Consequence if Not Corrected
Date	object (string)	datetime	If this variable is treated as a string, it is unable to perform time-based analysis like trend analysis, seasonal patterns, and time series modelling. It also makes it difficult to sort data by date.
City	object (string)	category (nominal)	This results in inefficient memory utilization and slower processing times during merging and aggregating. Then, the algorithms will be difficult to understand the inputs and treated as unstructured text due to the variable not representing a finite set of categories.
Country	object (string)	category (nominal)	This is similar to the results from City, as inefficient memory utilization and reduced the performance in analysis. Then, it fails to clearly display the categorical groups, which is critical for comparative analysis.

A1.6 Column with Corrected inferred dtype for Variable Classification purpose

Issue 1 refers to an incorrect date data type. **Before conversion**, the `Date` variable is stored as an object rather than in datetime format. **After conversion**, `pd.to\_datetime()` is used to convert the `Date` variable to datetime format to enable time series analysis of “year,” “month,” and

“day. No missing or invalid date values were detected. Hence, it indicates no significant conversion errors, and it can proceed for variable classification.

Issue 2 refers to the need for some improvements to the `City` and `Country` data types. **Before conversion**, the `City` and `Country` variables were stored as an object type rather than category type. To improve memory efficiency and better reflect their nominal categorical features, both variables have been converted to categorical types. **After conversion**, the `City` and `Country` variables were converted from `object` to `categorical` dtype successfully and can proceed for variable classification.

A2. Variable Classification

A2.1 Additional Variable Transformations

Issue 3 refers to negative temperature values. The `Temperature` variable contains negative values. Although negative temperatures are valid in some geographical regions, these values should be verified to ensure they are not unit inconsistencies or data entry errors such as Celsius and Kelvin. Therefore, we have created a `Temp\_Category` with 3 different thresholds such as `Hot`, `Cold`, and `Neutral` to support our data analysis.

Justification for Issue 3:

Although negative temperature values were identified, we decided to retain them because it may represent valid observations depending on geographic context. The descriptive statistics show the temperature values range from -10 to 40, which is consistent with common weather conditions in Celsius. The presence of negative values also confirms that the data is ****NOT**** recorded in Kelvin, because Kelvin is an absolute temperature scale.

Issue 4 refers to the lack of air quality classification. This dataset does not provide air quality classifications like “Good”, “Moderate”, and “Unhealthy” for pollutant variables such as `PM2.5` and `PM10`. It makes the air quality levels cannot be directly interpreted without defined thresholds. Hence, a categorical AQI variable was created from `PM2.5` and `PM10` values using standard thresholds to improve interpretability.

Justification for Issue 4:

The air quality categories `AQI\_Category` were derived using both `PM2.5` and `PM10` concentrations that meet the thresholds specified in the World Health Organization (WHO) Air Quality Guidelines. Then, each pollutant is first classified into different risk levels such as `Very Good`, `Good`, `Moderate`, `Unhealthy`, `Very Unhealthy`, `Hazardous` and the overall air quality category is determined based on the highest (worst case) level. Hence, this approach reflects real-world air quality assessment practices, where the most hazardous pollutant determines overall risk.

A2.2 Classification Table

Based on the output of A2.2, the variables in the dataset were classified according to their data type, meaning, and role in the analysis.

`Categorical` variables such as `City` and `Country`

Since the `Categorical` variables such as `City` and `Country` represent labels with no inherent order, both are treated as `nominal` identifiers. The `Date` variable is classified as a `temporal continuous` feature because it supports time series analysis.

Then, the environmental measurements such as `PM2.5`, `PM10`, `NO2`, `SO2`, `CO`, `O3`, `Temperature`, `Humidity`, and `Wind Speed` are classified as `continuous numerical` variables since they are measured on a continuous scale and can take a wide range of values. For these variables, mean or median is an appropriate descriptive statistic, while histograms or boxplots are suitable for visualizing its distributions.

Also, `Derived` variables are the variables created from existing variables to enhance interpretability and support in future data analysis. In this dataset, variables such as `Temp\_Category`, `AQI\_Level`, `PM25\_Level`, `PM10\_Level`, and `AQI\_Category` were generated from original measurements and classified as `ordinal` categorical because they represent ordered categories based on pollution intensity. These are best aggregated using frequency statistics and visualised using bar charts. Similarly, `Year`, `Month`, and `Day` are `discrete` numerical variables as it takes `integer` values within a fixed range and is derived from the `Date` variable.

In summary, this classification ensures that appropriate statistical methods and visualizations are applied to improve the accuracy and interpretability of the analysis.

A3 Missing Data Diagnosis

A3.1 Missing Data Pattern - Numerical summary

From the A3.1 output, the analysis of missing data shows that the proportion of missing values for the pollutant variables (`PM2.5`, `PM10`, `NO2`, `SO2`, `CO`, `O3`) ranges from approximately 4.8% to 5.3%, while remaining variables have no missing observations. This also implies that missing values are concentrated within a group of related variables rather than randomly distributed across the entire dataset. In this context, the pattern indicates that the missing data is related to the measurement of air quality indicators, or the data collection process compared to occur in completely at random. Therefore, further investigation is required to determine whether the underlying missing data mechanism is Missing Completely at Random (MCAR), Missing at Random (MAR), or Missing Not at Random (MNAR).

A3.2 Visualization - Missing Data Pattern

From the A3.2 output, the missing value heatmap shows that missing values are primarily concentrated in the air pollutant variables (`PM2.5`, `PM10`, `NO2`, `SO2`, `CO`, and `O3`), while other variables such as `Temperature`, `Humidity`, and `Wind Speed` have complete data. Also, the missing values appear as scattered gaps across different observations rather than forming large continuous blocks. In conclusion, the missing values do not follow any obvious pattern, and the pattern is more consistent with characteristics of **Missing at Random (MAR)**.

A3.3 Classify missingness (MCAR / MAR / MNAR)

The A3.3 result shows that the missing data pattern is different between variables and more concentrated in pollution measurements. Variables including 'City', 'Country', 'Date', 'Temperature', 'Humidity', and 'Wind Speed' have no missing values. However, variables such as 'SO2' and 'CO' have less than 5% missing values, whereas 'PM2.5', 'PM10', 'NO2', and 'O3' have more than 5% missing values. The missing values are distributed across the data rather than making identifiable or continuous blocks. Then, the pattern does not indicate **Missing Completely at Random (MCAR)**, as the missing values are not randomly distributed across the dataset. Moreover, it is also unlikely to be **Missing Not at Random (MNAR)**, as the probability of missingness does not appear to depend on the unobserved values (Mazraeh, 2025). In addition, the missing values may be influenced by observable factors such as environmental conditions or sensor availability rather than occurring completely at random.

Thus, the missingness is most consistent with **Missing at Random (MAR)**, as it is concentrated within specific pollutant variables rather than occurring uniformly across all variables as shown in the previous heatmap.

A3.4 Compare handling strategies - for `PM2.5`

From the A3.4 output, a comparative handling strategies analysis is conducted for a particular variable `PM2.5` on the cleaned dataset (df_clean) to ensure consistency after preprocessing. The listwise deletion and median imputation have been conducted to investigate appropriate methods for handling missing data. The listwise deletion reduces the dataset size and may cause bias when missingness is not completely random, whereas median imputation retains all observations and provides a reliable estimate of central tendency (GeeksforGeeks, 2025).

After conducting the analysis, it discovered that both methods produce similar mean values, but median imputation had a slightly smaller standard deviation. This is because imputing using the median reduces variability in the dataset by replacing them with a central value. Therefore, median imputation is recommended for the remaining missing values since it retains all data while producing consistent results.

Justification for MCAR, MAR, and MNAR:

The difference between MCAR, MAR, and MNAR is stated as below:

- Missing Completely at Random (MCAR) occurs when the probability of missingness is independent of both observed and unobserved data.
- Missing at Random (MAR) occurs when missingness depends only on observed variables.
- Missing Not at Random (MNAR) occurs when missingness depends on the unobserved variables or the missing values themselves.

Missing values must be handled appropriately, and the ideal handling approach can be determined by the pattern of missingness. If the data is MCAR, listwise elimination offers unbiased outcomes. However, if the data is MAR or MNAR, imputation methods are preferred to eliminate bias and retain statistical validity. For instance, mean imputation is more suitable when the data is approximately normally distributed without significant outliers, while median imputation is preferable when the data is skewed or contains outliers, as it is less affected by the extreme values.

A3.5 Data Cleaning

According to the summary table of the A3.5 output, the structural quality of the original dataset is generally good and suitable for statistical analysis. This is because the dataset contains a sufficient number of observations, and the variables are clearly defined. Most numerical variables have the correct data types, and key contextual variables (such as 'City' and 'Country') are complete. However, there are several quality issues in the dataset, such as duplicate rows, missing values, and inconsistent variable formats; therefore, data preprocessing is required before conducting further analysis.

A3.5.1 Key Data Quality Issues

Issue 5 refers to the duplicate and near-duplicate rows. The dataset contains duplicate and near-duplicate observations, which result in biased statistical summaries by overestimating some records. Hence, it is necessary to remove duplicate rows to ensure that each observation contributes equally to the analysis. **After cleaning the duplicate**, a sample of duplicate rows is given to provide visual evidence of repeated observations within the dataset. It displays both the original and duplicated entries to ensure that the rows are identical across all variables. Thus, the decision to delete duplicates is rational because their presence could potentially affect statistical summaries by misrepresenting certain observations. Also, a table shows that duplication detection was conducted following standardization to ensure the formatting inconsistencies did not misrepresent similar entries.

Issue 6 refers to missing values. There are several pollution variables ('PM2.5', 'PM10', 'NO2', 'SO2', 'CO', and 'O3') have around 5% missing values and appears to be scattered, which is

consistent with Missing at Random (MAR). If not managed appropriately, it could lead to bias. Hence, median imputation is preferred due to the skewed distribution of environmental data. Then, missing values in pollutant variables were handled simultaneously using median imputation. This approach maintains consistency across variables and is suitable for the skewed distribution of environmental data.

Issue 7 refers to inconsistent variable formatting in variables such as `Country` and `City`, including differences in type, capitalization, spacing, and naming conventions (i.e., "USA" vs "usa"). This issue can lead to inaccurate grouping and aggregation. Hence, standardization is required prior to analysis. **After standardization**, the `City` variable was standardized by removing whitespace and converting text to lowercase. It reduces the number of unique categories and more consistent grouping.

A4 Data Quality Narrative

After data cleansing and validation, the dataset is normally sufficient for statistical analysis. An initial inspection reveals that the dataset has various quality issues. There were two major issues: missing values and data inconsistencies (i.e., duplicate and data types issues) that did not align. The first issue is **missing values** that are particularly significant because it could cause bias or make statistics less accurate. It was handled by comparing listwise deletion and median imputation. The median imputation was used as less affected by outliers to maintain its reliability. Then, the second issue is **data inconsistencies** such as duplicates were removed with proper handling. As data types were incomplete, especially for variables like `Date`, `City`, and `Country`. We standardized these by converting the `Date` into a `datetime` format and changing the `City` and `Country` as `categorical` variables to make the data more consistent and easier to analyze. Also, additional variables like `Temp\_Category` and `AQI\_Category` were created to better interpret the data.

The dataset has become more well-structured and maintains its data quality. However, certain decisions might have an impact on the future data analysis. For instance, the method of imputation chosen might affect summary statistics and variability. Also, the additional categories variables that are derived by continuous variables such as temperature and air quality levels will cause it to be less detailed. In general, the `df\_clean` dataset is clean, consistent, and reliable sufficient to conduct a valid statistical analysis.

A4.1 Overview of the Final Cleaned Dataset: `df\_clean`

From the A4.1 output, it can be claimed that the dataset `df\_clean` contains **19955 rows and 20 columns** is cleaned and ready for descriptive statistical analysis in Task B.

Task B – Descriptive Statistics and Distribution Shape

B1. Univariate Summaries

This section examines key descriptive statistics for all numerical variables, including measures of central tendency, dispersion, and distribution shape. The variables analysed are PM2.5, PM10, NO₂, SO₂, CO, O₃, Temperature, Humidity, and Wind Speed.

Key Findings

- Central Tendency
 - For all variables, the mean and median are very close (e.g., PM2.5: mean = 77.76, median = 78.35), indicating approximately symmetric distributions. This suggests that extreme values do not heavily distort the mean, making both measures reliable indicators of central tendency.
- Spread
 - Pollution-related variables such as PM2.5, PM10, and O₃ exhibit relatively large standard deviations and IQRs, indicating substantial variability in pollution levels. In contrast, CO and Wind Speed show smaller dispersion, suggesting more stable measurements.
- Skewness
 - All variables have skewness values very close to zero (approximately -0.02 to 0.02), indicating minimal skewness and near-symmetric distributions. No variable shows a strong right or left skew.
- Kurtosis
 - All variables display negative excess kurtosis (≈ -1), indicating platykurtic distributions. This means the distributions are flatter than normal with lighter tails, suggesting fewer extreme values than expected under a normal distribution.
- Outliers and Distribution Shape
 - The combination of near-zero skewness and negative kurtosis suggests that the data is relatively evenly distributed without strong peaks or heavy tails. While high values exist (e.g., PM10 up to 200), they are not extreme relative to the overall distribution.

Justification of Summary Measures

Given the near-symmetric distributions, the mean is recommended as the primary measure of centre for all variables, as it incorporates all observations and is not significantly affected by skewness.

However, the median remains a robust alternative, particularly in environmental data where occasional extreme values may occur.

For dispersion:

- The IQR is preferred for variables with higher variability (PM2.5, PM10, O₃), as it is robust to extreme values.
- The standard deviation is appropriate for more stable variables (CO, Wind Speed), where variability is low and evenly distributed.

From a practical perspective, pollution variables naturally fluctuate due to environmental factors, so using robust measures like the IQR provides a more reliable representation of spread.

B2. Outlier Investigation

Outliers were assessed using two methods:

1. IQR method ($1.5 \times \text{IQR}$ rule)
2. Z-score method ($|z| > 3$)

Key Findings

No outliers were detected for PM2.5 and PM10 using either method. However, inspection of raw values shows relatively high observations:

- PM2.5 reaches approximately 150
- PM10 reaches approximately 200

These values are higher than the mean but:

- Fall within the IQR-based bounds
- Do not exceed the Z-score threshold

This indicates that the dataset has a wide but well-distributed range, rather than isolated extreme spikes.

Theoretical Comparison of Methods

The IQR method is non-parametric and does not assume any distribution. It identifies outliers based on the spread of the middle 50% of the data, making it robust to skewness and extreme values.

The Z-score method assumes an approximately normal distribution and defines outliers relative to the mean and standard deviation. It is therefore sensitive to skewness and extreme values, which can distort both the mean and standard deviation.

When Z-score Can Fail

The Z-score method may fail when:

- The data is skewed, causing the mean to shift
- The distribution has heavy tails

- The data is non-normal

In such cases, true outliers may not exceed the ± 3 threshold and may go undetected.

Why Outliers Should Not Be Automatically Removed

Outliers should not be removed automatically because:

- They may represent real-world phenomena (e.g., pollution spikes)
- Removing them can bias analysis and reduce variability
- They may provide important domain insights

In this dataset, high pollution values are plausible observations, not errors and therefore should be retained.

B3. Grouped Comparison

PM2.5 levels were compared across AQI categories: Good, Moderate, Unhealthy, Very Unhealthy and Hazardous.

Key Findings

There is a clear monotonic increase in PM2.5 levels across AQI categories:

- Median increases from 9.36 (Good) to 114.88 (Hazardous)

This indicates progressively worsening air quality as category severity increases.

Spread

- Lower categories (Good, Moderate) show small IQRs ($\approx 4.75 - 8.46$), indicating stable conditions
- Higher categories (Unhealthy to Hazardous) show much larger IQRs ($\approx 19.96 - 50.78$), indicating greater variability

Standard deviation follows the same increasing pattern.

Shape and Outliers

Distributions within each category appear approximately symmetric, consistent with earlier findings. Very few outliers are visible in the boxplot, supporting the conclusion that extreme values are limited.

Justification for Using Median

The median is preferred over the mean for group comparison because:

- Variability differs significantly across groups
- Higher AQI categories show much wider spread, making means more sensitive to extreme values

- Median provides a robust measure of central tendency

Using the median ensures fair comparison across groups. For example, the steady increase in median PM2.5 clearly reflects worsening pollution without distortion from variability.

B4. Correlation and Covariance

Key Findings

The Pearson correlation matrix shows that all pairwise correlations are extremely weak, with values between approximately -0.02 and 0.02.

- Strongest: SO₂ and CO ($r \approx 0.024$)
- Weakest: Temperature and PM2.5 ($r \approx -0.021$)

The scatter plot of SO₂ vs CO shows a random cloud of points, confirming the absence of a linear relationship.

Spearman correlation results are nearly identical, indicating no strong monotonic relationships.

Mathematical Definition

The Pearson correlation coefficient is defined as:

$$r = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y}$$

It measures the strength and direction of a linear relationship between two variables.

Critical Assumptions

- The relationship is linear
- The data is not heavily influenced by outliers

Properties

Correlation is dimensionless because covariance has units of (unit_x × unit_y), while the standard deviations have units of unit_x and unit_y respectively. Dividing covariance by the product of standard deviations cancels all units, allowing comparison across variables measured in different scales.

Correlation is also symmetric:

- $\text{Corr}(\text{PM}_{2.5}, \text{PM}_{10}) = 0.0162$
- $\text{Corr}(\text{PM}_{10}, \text{PM}_{2.5}) = 0.0162$

Spearman Correlation

Spearman correlation is based on rank order and measures monotonic relationships.

It is preferred when:

- Relationships are non-linear
- Data contains outliers
- Data is non-normal

In this dataset, Spearman results confirm that there is no evidence of strong linear or monotonic relationships between variables, with no strong monotonic trends.

Conclusion

The dataset exhibits:

- Near-symmetric distributions with minimal skewness
- Moderate variability, especially in pollution variables
- No statistically significant outliers under standard definitions
- Clear group differences in PM_{2.5} across AQI categories
- Minimal correlation between variables

These findings suggest that:

- Air quality variables vary widely but in a structured and consistent manner
- AQI categories effectively capture both level and variability of pollution
- Relationships between variables are weak, indicating that factors may operate independently or require more complex modelling to uncover interactions

Task C – Visualisation Audit and Storytelling

C1. Required Plots and Statistical Justification

In this section, we transition from numerical summaries to visual arguments. A visualisation is not merely a picture; it is a structured argument that can either clarify or mislead based on design choices. We focus on PM2.5 as our primary variable of interest.

Histogram Bin Count Justification

To accurately capture the distribution of PM2.5 levels across the 13,595 observations in our dataset, we evaluated two mathematical frameworks for determining the optimal number of bins:

1. Sturges' Rule:

- Formula: $k = \log_2(n) + 1$
- Application: $k = \log_2(13,595) + 1 \approx 13.7 + 1 = 14.7$
- Sturges' rule assumes a normal distribution and is generally better suited for smaller datasets.

$$k = \lceil \log_2(13,595) + 1 \rceil \approx \lceil 13.73 + 1 \rceil = 15 \text{ bins}$$

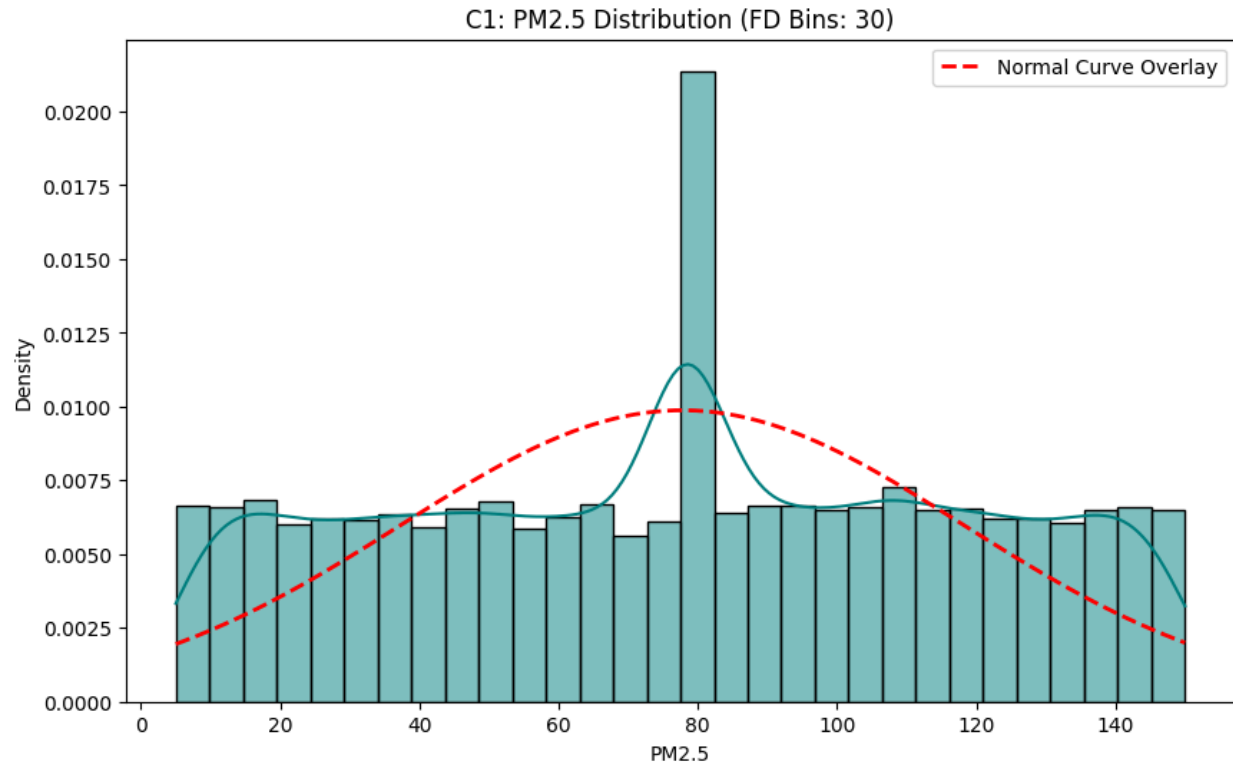
2. Freedman–Diaconis Rule:

- Formula: Bin Width = $2 * \text{IQR} / (n^{1/3})$
- Calculation: With an IQR of 50.78, Bin Width = $(2 * 50.78) / (13,595^{1/3}) \approx 101.56 / 23.87 \approx 4.25$
- This width suggests approximately 35 bins over the data range.

$$\text{Bin Width} = 2 \times \frac{\text{IQR}}{n^{1/3}} = 2 \times \frac{50.78}{13,595^{1/3}} \approx \frac{101.56}{23.87} \approx 4.25$$

$$\text{Bin Count} = \frac{\text{Range}}{\text{Bin Width}} \approx \frac{150}{4.25} \approx 35 \text{ bins}$$

We selected the **FD rule (resulting in 30 bins)**. Because air quality data is inherently non-normal and heavily skewed, the FD rule which utilizes the **Interquartile Range (IQR)** far more resilient to outliers than Sturges' rule.



Ensure it includes the FD-calculated bins, a KDE (Kernel Density Estimate) line, and the red dashed Normal Distribution overlay for comparison

Justification of choice:

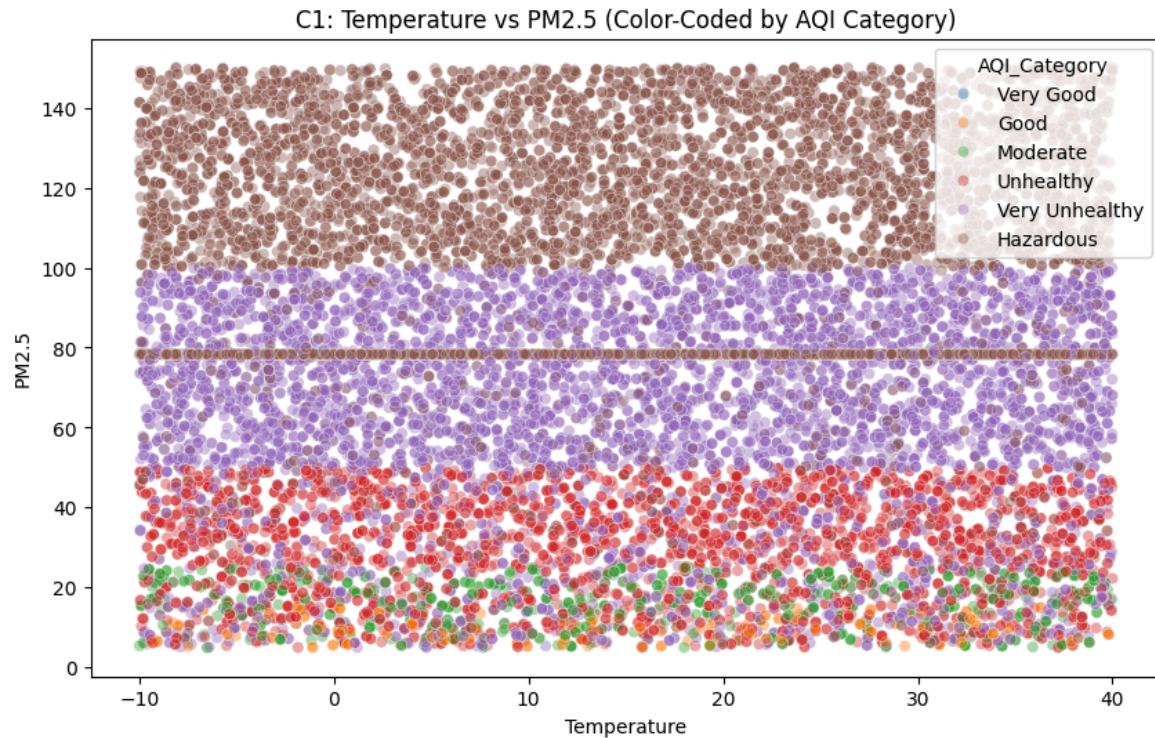
Although Sturges' rule is simple and commonly used, it may not be the best option for large datasets because it can over-smooth the data and hide important variations, such as sudden spikes or changes in pollution levels. This can cause the histogram to appear less detailed than the actual data distribution. Therefore, the Freedman–Diaconis rule was selected because it takes into account the spread of the data by using the interquartile range (IQR). This makes it more effective in representing the true pattern and variability of air quality data.

Selecting an inappropriate bin width can significantly affect how readers interpret the results. If too few bins are used, the histogram may make the data look more uniform and less variable than it really is. On the other hand, if too many bins are used, the graph may become crowded with unnecessary visual noise, making it harder to identify the overall trend. For this reason, the Freedman–Diaconis rule provides a more balanced and reliable visualization.

Correlation & Variance Observations

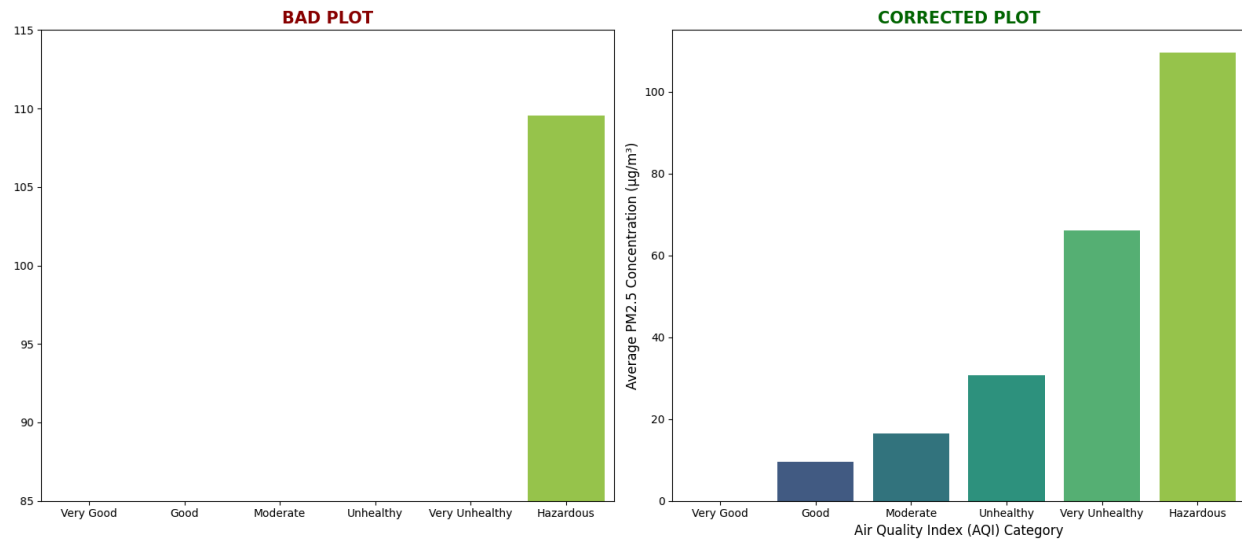
Analysis via scatter and box plots indicates that pollution is not evenly distributed across categories.

- **The Volatility Factor:** As air quality moves from "Good" to "Unhealthy," the **variance** (the spread of data) expands. This proves that hazardous air quality is not just higher in concentration, but significantly more unpredictable.



C2. The Deliberately Bad Plot

The image below contrasts a deliberately misleading bar chart with its honest counterpart. The "Bad Plot" is designed to create a sense of environmental crisis through deceptive scaling, while the "Corrected Plot" provides a proportional view of the PM2.5 averages across standard AQI categories.



Data science carries a moral obligation to represent truth, yet visualizations are often weaponized to serve alarmist or biased narratives. To explore the boundaries of this responsibility, we developed a "Bad Plot" designed to mislead the viewer through the following specific anti-patterns:

- **Truncated Vertical Scaling (The Cut-Off Axis):** The Y-axis baseline was intentionally set to **85** instead of **0**. This exploits how the human brain perceives magnitude; by removing the bottom 85 units, a small numerical difference is visually magnified into a massive gap, making certain categories look five times larger than they actually are.
- **Contextual Stripping (Missing Labels and Units):** All axis titles, specific category names, and units of measurement $\mu\text{g}/\text{m}^3$ were removed. This strips the data of its frame of reference, forcing the viewer to rely on the distorted "shapes" of the bars rather than factual, verifiable evidence.

Why These Anti-Patterns Are Scientifically Wrong

The use of Truncated Vertical Scaling is fundamentally dishonest because it violates Proportional Integrity. Since we judge the value of a bar by its total length, cutting the axis creates a "false alarm" state where a 20% increase in pollution looks like a 500% spike. Similarly, Contextual Stripping is a violation of the principle of Transparency. Without units or labels, the data becomes an abstract shape that cannot be checked against global health standards, leaving the audience vulnerable to emotional manipulation. In contrast, the Corrected

Plot restores the zero-baseline and clear metadata, ensuring that the visual area of every bar is perfectly proportional to its real mathematical value. Ethical visualization must always prioritize the integrity of the data over the visual drama of the presentation.

C3. The Story of Urban Volatility

"How do extreme environmental temperatures influence the volatility of PM2.5 concentrations, and are our standard air quality categories effectively capturing the risk of exposure during these thermal events?"

Analysis of the [global_air_quality.csv](#) dataset reveals that pollution is not a static threat; it is a dynamic phenomenon heavily dictated by the surrounding environment. By examining the **C1 Histogram**, we immediately see that air quality does not follow a "Normal Distribution." While the theoretical curve suggests a balanced spread, the actual data is **heavily right-skewed** with a significant "long tail." This statistical profile confirms that extreme pollution events are not mere statistical flukes, but recurring systemic hazards that occur more frequently than standard models predict.

When we cross-reference this with the **Scatter Plot (Temperature vs. PM2.5)**, a clear environmental correlation emerges. We observe that the highest concentrations of PM2.5 often classified as "Unhealthy" or "Hazardous" which tend to cluster at specific temperature extremes. In lower temperatures (below 0°C), this is likely caused by **"thermal inversions,"** where a layer of cold air near the ground traps pollutants that would otherwise disperse. Conversely, in extreme heat (above 35°C), stagnant air masses and chemical reactions often lead to the formation of secondary particulates.

The **Ordered Box Plot** further supports this narrative by showing that as we transition from "Good" to "Unhealthy" categories, the **variance** (the spread of the data) expands dramatically. This indicates that once air quality begins to degrade, it becomes highly volatile and unpredictable. For a student or a citizen, this means that on a "Moderate" day, you can predict your exposure with relative certainty. However, on an "Unhealthy" day, pollution levels can spike to dangerous extremes within minutes, representing a failure in the stability of current categorical warnings.

Data Limitations

A primary limitation of this analysis is the lack of **Temporal Granularity**. The dataset provides a single snapshot of PM2.5 and Temperature, likely as a daily average. However, air quality is notoriously sensitive to "Rush Hour" traffic patterns and domestic heating cycles. Without hourly timestamps, we cannot definitively determine whether the observed temperature correlations are driven solely by natural atmospheric traps or by human behavior, such as increased vehicle idling or wood-burning during temperature extremes.

Conclusion

The data demonstrates that air quality risks are significantly amplified by temperature extremes, which act as catalysts for pollutant stagnation. Therefore, municipal health alerts should not be purely reactive to current PM2.5 levels. Instead, cities should implement **"Weather-Triggered Preemptive Alerts."** When local forecasts predict extreme temperature windows, public transport subsidies should be activated and industrial emissions capped *before* the PM2.5 spike occurs, effectively mitigating the environmental "trap" before it closes on the public.

Task D – Probability Theory and Bayesian Reasoning

This task applies probability theory and Bayesian reasoning to the cleaned air quality dataset (df_clean). It is divided into three sections:

1. D1 uses analytical probability to compute and verify probabilities from observed data
2. D2 uses Monte Carlo simulation to estimate probabilities through repeated experiments
3. D3 applies Bayesian updating to revise probabilities based on observed evidence

D1. Analytical Probability

Variable Selection and justification

Two variables were selected from the probability analysis:

Role	Variable	Values
A (categorical)	AQI_Category (collapsed)	A1 = Unhealthy A2 = Acceptable
B (binary)	High_NO2	B=1 if NO2 > median B=0 otherwise

The variable

AQI_Category is derived from PM2.5 and PM10, which measure tiny particles in the air. These particles are commonly used to assess the overall air quality. In contrast, NO2 is a gaseous pollutant mainly produced by combustion processes such as traffic and industrial activity and is not included in the AQI calculation.

Therefore, these two variables represent different types of air pollution, making them useful for probability analysis. The aim is to examine whether higher NO₂ levels are associated with worse AQI categories.

A 2x2 contingency table was constructed to examine the relationship between AQI category and NO2 level. The table below shows how the observations are distributed across the two variables. These counts are then used to compute the joint and conditional probabilities in the following section.

AQI Category (A)	Low NO2 (B = 0)	High NO2 (B = 1)	Total
Acceptable	450	415	865
Unhealthy	10284	8806	19090
Total	10734	9221	19955

The following probabilities were calculated:

Probability	Value	Description
$P(A = \text{Unhealthy})$	0.9567	The majority of observations are classified as Unhealthy.
$P(B = 1)$	0.4621	About 46.2% of days have above-median NO2 levels.
$P(A \cap B)$	0.4413	44.1% of observations are both Unhealthy and have high NO2.
$P(B = 1 \mid A = \text{Unhealthy})$	0.4613	High NO2 occurs at a similar rate on Unhealthy days.
$P(B = 1 \mid A = \text{Acceptable})$	0.4798	High NO2 also occurs at a similar rate on Acceptable days.
$P(A = \text{Unhealthy} \mid B = 1)$	0.9550	Given high NO2, most observations still fall into the dominant Unhealthy AQI category.

Key Insights:
The

conditional probabilities, $P(B = 1 \mid \text{Unhealthy}) \approx 0.461$ and $P(B = 1 \mid \text{Acceptable}) \approx 0.480$ are almost similar, suggesting that a high NO2 does not strongly distinguish between AQI categories.

Bayes' Theorem Verification

Bayes' theorem was used to verify the conditional probability:

$$P(A | B) = \frac{P(B | A)P(A)}{P(B)}$$

The probability $P(A = \text{Unhealthy} | B = 1)$ was calculated in two ways:

1. Direct computation from the contingency table
2. Using the Bayes' theorem

Key Findings:

Both approaches produced similar results:

- Direct calculation: $P(A=\text{Unhealthy} \mid B=\text{High NO}_2) = 0.95499404$
- Via Bayes theorem: $= 0.95499404$
- Difference: $= 1.1102230246251565e-16$

Interpretation:

This shows that the probability calculations are consistent and confirms that the relationship defined by Bayes' theorem is correctly reflected in the data. It also demonstrates how Bayes' theorem can be used to reverse conditional probabilities, in this case estimating the likelihood of Unhealthy air quality given high NO₂ levels.

Independence Test

Two events A and B are independent if: $P(A \cap B) = P(A) * P(B)$

To assess this relationship, the observed joint probability was compared with the expected probability assuming independence.

From the results:

- Observed joint probability ≈ 0.4413
- Expected probability ≈ 0.4421
- Chi-square test: $p = 0.286 (> 0.05)$

The observed and expected probabilities show a really close value, which means that there is little difference between the observed and expected outcomes

Chi-square test result: $0.286p=0.286$ (greater than 0.05)

Interpretation

Since the p-value is greater than 0.05, we do not have enough evidence to say that the variables are related and we fail to reject the null hypothesis of independence. In simple terms, this means that NO₂ levels and AQI category appear to be independent, and higher NO₂ does not necessarily correspond to worse AQI categories in this dataset.

Effect Size

To assess the strength of any association, Relative Risk and Odds Ratio were calculated:

Relative Risk (RR) = 0.961

Odds Ratio (OR) = 0.928

Both measures are close to 1, indicating a very weak association between AQI category and NO2 levels. This aligns with the independence test results and confirms that any relationship is negligible in practical terms.

D2. Monte Carlo Simulation

Monte Carlo simulation is a technique used to estimate probabilities by repeating a random experiment multiple times and observing the outcomes. It is based on the Law of Large Numbers, which states that as the number of trials increases, the simulated results converge to the true probability.

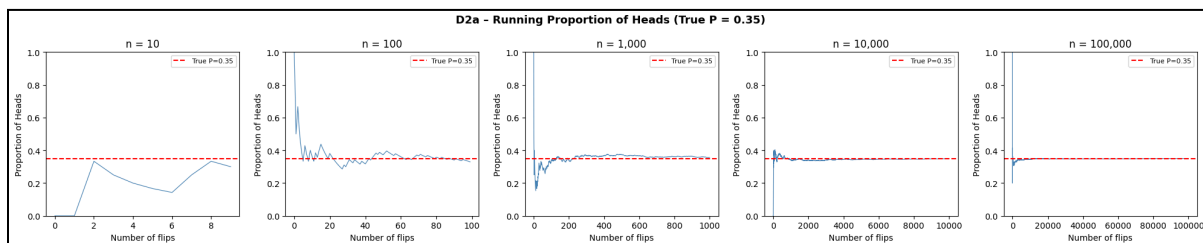
All simulations follow four steps:

1. Define the random process
2. Write a function for one trial
3. Choose the number of repetitions
4. Collect results in a loop.

A random seed (`np.random.seed(42)`) was set before all simulations. This ensures that the same random numbers are generated each time the notebook is run, allowing results to remain consistent.

D2a. Biased Coin with $P(\text{Heads}) = 0.35$

A biased coin with probability of heads equal to 0.35 was simulated using repeated Bernoulli trials. The running proportion of heads was tracked for increasing sample sizes: $n = 10, 100, 1000, 10000$, and 100000 .



Final proportion of heads after all flips:

```
n =      10: 0.3000 (true P = 0.35)
n =     100: 0.2500 (true P = 0.35)
n =    1,000: 0.3790 (true P = 0.35)
n =   10,000: 0.3519 (true P = 0.35)
n =  100,000: 0.3511 (true P = 0.35)
```

Key Findings:

1. At small sample sizes ($n = 10$), the running proportion fluctuates significantly due to randomness.
2. As the number of trials increases, the proportion becomes more stable.
3. At $n = 100,000$, the simulated proportion closely converges to the true probability of 0.35, with only minimal deviation.

Interpretation:

This behaviour illustrates the Law of Large Numbers, where increasing the number of observations reduces the effect of random variation, causing the sample proportion to approach the true probability.

Mathematically, the variance of the sample proportion is:

$$\frac{P(1-P)}{n}$$

As n increases, the variance decreases, which explains why the estimate becomes more stable and concentrated around the true value. This principle forms the foundation of Monte Carlo simulation methods, where repeated random sampling is used to approximate unknown probabilities.

D2b. Bootstrap Estimate of Conditional Probability

The conditional probability $P(B=1 | A=\text{Unhealthy})$ was estimated using **50,000 bootstrap samples** drawn from the Unhealthy subset of `df_clean`

Simulation Workflow

The Monte Carlo procedure follows the required four steps:

1. Define the process
 - a. Estimate the proportion of high NO_2 days within the Unhealthy AQI group.
2. Single trial function
 - a. Randomly sample rows from the Unhealthy subset
 - b. Use sampling with replacement
 - c. Compute the proportion of observations where $B=1$
3. Number of repetitions
 - a. The process is repeated 50,000 times
4. Collect results
 - a. Store each estimated probability to form a distribution of bootstrap estimates

Role of Sampling with Replacement:

Sampling with replacement ensures that each resample is slightly different from the original data. Without replacement, every sample would be identical, and no variability could be observed.

This variability is essential because it allows the bootstrap method to estimate uncertainty in the probability.

Key findings:

```
Number of Unhealthy rows to resample from: 19,090
Running 50,000 bootstrap samples...

Analytical value (from D1): 0.4613
Bootstrap mean:             0.4613
Bootstrap standard error:   0.0036
95% Bootstrap CI:          [0.4543, 0.4684]
|Bootstrap - Analytical|:   0.000006
```

From the simulation output, the key findings are:

1. Analytical value (from D1): 0.4613
2. Bootstrap mean: 0.4613
3. Standard error: 0.0036
4. 95% confidence interval: [0.4543, 0.4684]
5. Absolute difference: 0.000006

Interpretation

The bootstrap mean is almost identical to the analytical value, with an extremely small difference. This shows that the simulation accurately reproduces the true probability.

The standard error is low, indicating that the estimates are stable across repeated samples. The confidence interval is narrow and contains the analytical value, confirming that the estimate is reliable.

Overall, this shows that:

1. The Monte Carlo simulation provides an accurate estimate
2. Sampling variability is small
3. The result is robust and consistent with the analytical calculation

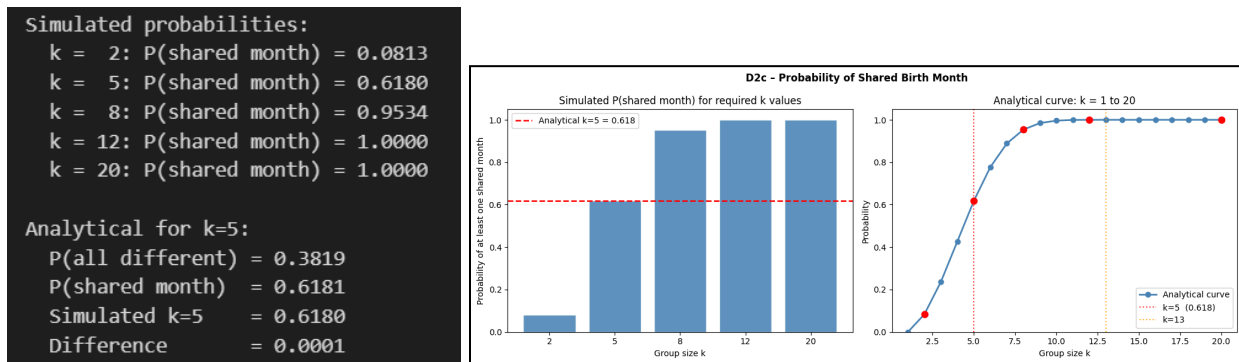
This is also consistent with the Law of Large Numbers, where increasing the number of simulations leads to more stable and accurate estimates.

D2c. Birthday Month Problem

This simulation estimates the probability that at least two people in a group share the same birth month, assuming there are 12 months in a year.

The analysis was conducted for group sizes $k = 2, 5, 8, 12, 20$

In each trial, k individuals are randomly assigned a birth month (from 1 to 12). The trial returns a value of 1 if at least one repeated month is detected within the group, and 0 otherwise. The final probability is obtained by averaging the results across all simulations.



Analytical Chance ($k = 5$)

The exact probability is derived using the complement rule:

- $P(\text{at least one shared month}) = 1 - P(\text{all different})$
- $P(\text{all different}) = \frac{12}{12} \times \frac{11}{12} \times \frac{10}{12} \times \frac{9}{12} \times \frac{8}{12} \approx 0.3819$
- $P(\text{shared month}) \approx 1 - 0.3819 = 0.6181$

Result Comparison:

- Analytical: 0.6181
- Simulated: 0.6180
- Difference: 0.0001

Interpretation

The simulated result closely matches the analytical value, with only a very small difference due to random sampling variation. This confirms that the Monte Carlo simulation provides a reliable approximation when a sufficiently large number of trials is used.

Why the Probability Increases Quickly

The probability increases rapidly as group size increases because each new person must avoid all previously chosen months.

1. Person 1: no restriction = 12/12
2. Person 2: avoid 1 month = 11/12
3. Person 3: avoid 2 months = 10/12
4. and so on

These probabilities multiply together, which causes the probability of all distinct months to decrease quickly. As a result, the likelihood of at least one shared month increases sharply with group size. By $k=12$, the probability is already extremely high. At $k=13$, a shared month is guaranteed by the Pigeonhole Principle, since there are more people than available months.

D3. Bayesian Update on Real Data

Bayesian inference updates an initial probability (prior belief) using observed evidence to obtain an updated probability (posterior). The relationship is defined by:

$$P(H|E) = \frac{P(E|H) \times P(H)}{P(E)}$$

where:

H: event that a day is classified as Hot (Temperature > 28°C)

E: event that humidity exceeds 75%

This analysis investigates whether high humidity provides meaningful information about whether a day is hot.

Dataset Summary

The dataset contains 19,955 observations. After defining the events:

- Hot days ($H = 1$): 4,763 (23.9%)
- Non-Hot days ($H = 0$): 15,192 (76.1%)
- High humidity days ($E = 1$): 5,418 (27.2%)
- High humidity and Hot: 1,288
- High humidity and not Hot: 4,130

These values form the basis for computing the prior, likelihood, and posterior probabilities.

Bayesian Results

- The computed probabilities are:
- Prior $P(H) = 0.2387$
- Likelihood $P(E|H) = 0.2704$
- Likelihood $P(E|\neg H) = 0.2719$
- Marginal $P(E) = 0.2715$
- Posterior $P(H|E) = 0.2377$
- Likelihood ratio = 0.995

Interpretation of Results

The prior probability indicates that approximately 23.9% of all days are classified as Hot. After observing high humidity conditions, this probability changes only slightly to 23.8%. The very small difference between the prior and posterior (-0.001) shows that humidity provides almost no additional information about whether a day is Hot.

This is further supported by the likelihood ratio of approximately 1.0, which indicates that high humidity is almost equally likely on both Hot and non-Hot days. In other words, humidity does not meaningfully distinguish between temperature categories in this dataset.

Meaning of the Bayesian Update

1. The prior (23.9%) represents the baseline likelihood of a Hot day without any additional information.
2. The likelihood values show how humidity behaves within each temperature group, but both are nearly identical.
3. The posterior (23.8%) shows that even after observing high humidity, our belief about Hot days remains almost unchanged.

This indicates that humidity is a weak predictor of temperature category in this dataset.

Sensitivity Analysis

A sensitivity analysis was conducted by varying the prior probability from 0.1 to 0.9 while keeping the likelihood fixed.

Key observations:

1. The posterior changes almost linearly with the prior.
2. The difference between prior and posterior remains extremely small across all values.
3. The maximum change is only about ± 0.0013 .

This shows that the posterior is not strongly influenced by the data (likelihood), meaning the evidence has low discriminatory power.

Interpretation of Prior vs Data Influence

In Bayesian terms, this result indicates that the prior dominates the inference because the likelihood ratio is approximately 1.

When $P(E|H) \approx P(E|\neg H)$, the evidence provides little to no update to the prior belief. As a result, the posterior remains almost identical to the prior. This explains why humidity does not meaningfully shift the probability of a Hot day in this dataset.

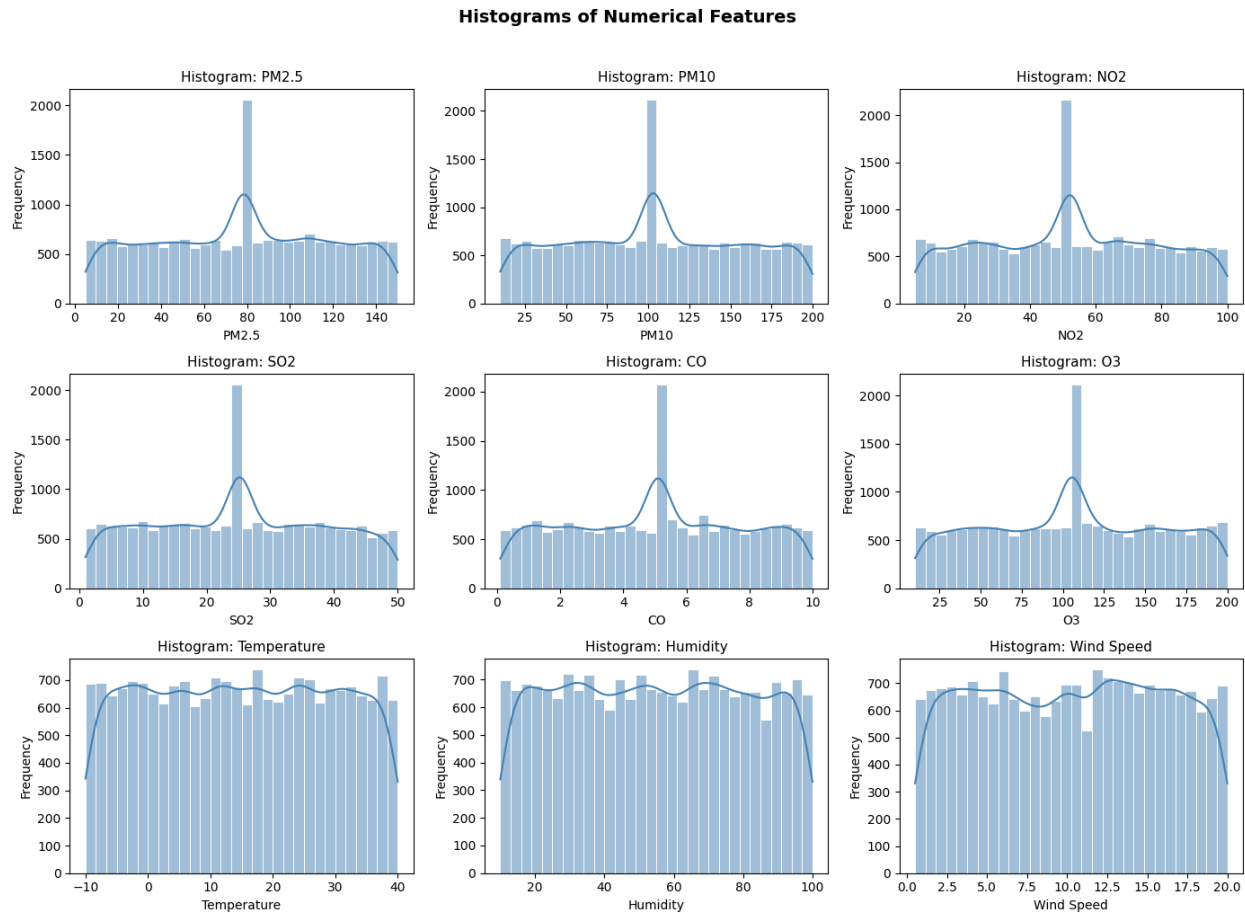
Conclusion

The Bayesian analysis demonstrates that humidity above 75% does not significantly influence the probability of a day being classified as Hot. The posterior probability remains nearly identical to the prior, confirming that temperature category and humidity are weakly related in this dataset. This suggests that temperature classification is likely driven by other meteorological or seasonal factors rather than humidity alone.

Task E – Statistical Distributions and Model Fitting

E1. Normality Assessment

A normality assessment is conducted for numerical features using descriptive statistics, visual diagnostics (histograms and Q-Q plots), and the Shapiro–Wilk test.



Variable Selection

- **Approximately Normal (NO_2):** Selected due to near-perfect symmetry (Skewness = 0.000) and the lowest platykurtic deviation (Excess Kurtosis = -1.032).
- **Clearly Non-Normal ($\text{PM}_{2.5}$):** Selected due to a significant central spike and a high platykurtic deviation (Excess Kurtosis = -1.059), indicating strong departure from a Gaussian curve.

Visual Diagnostics and Test Results

How to read the Q-Q plot:

Axes meaning:

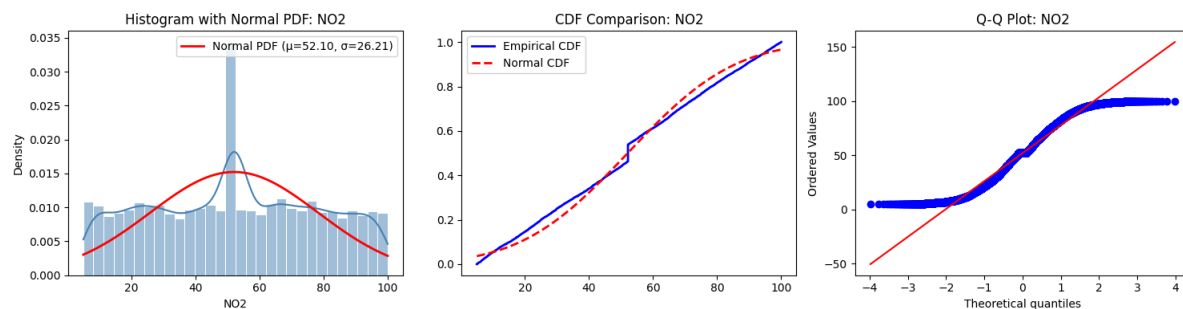
- X-axis (Theoretical Quantiles): Values expected if the data followed a perfect normal distribution.
- Y-axis (Sample Quantiles): Ordered values from the actual dataset.

Interpreting deviations from the diagonal line:

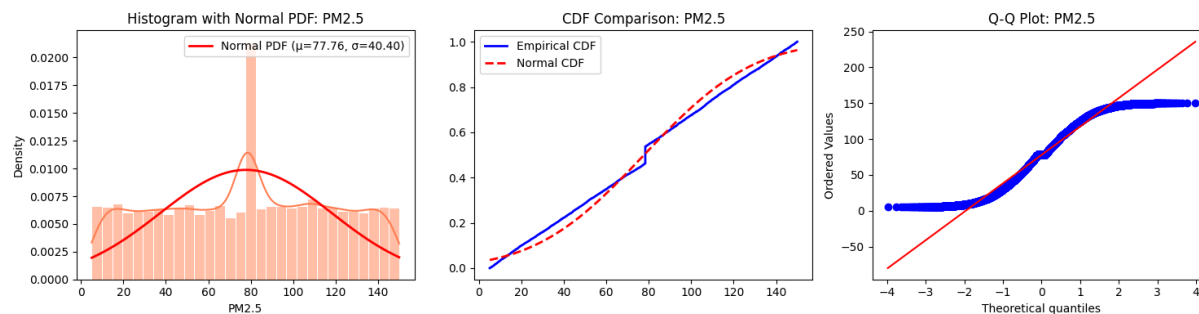
- Right-skew: Points bend upward in the right tail, showing a longer right tail than normal.
- Left-skew: Points bend downward in the left tail, showing a longer left tail than normal.
- Heavy tails: An S-shaped curve where both tails rise above the line, indicating more extreme values than normal.
- Light tails: An inverted S-shape where both tails fall below the line, indicating fewer extremes than normal.

Shapiro-Wilk Test:

- H_0 : Data is normally distributed
- H_1 : Data is not normally distributed
- Decision rule: Reject H_0 if p-value < 0.05



While NO₂ appears visually symmetric, the Shapiro-Wilk test yielded a p-value of 0.0000 ($W = 0.9662$), leading to formal rejection of the null hypothesis (H_0).



Also for PM2.5, the Shapiro-Wilk test yielded a p-value of 0.0000 ($W = 0.9641$), leading to formal rejection of the null hypothesis (H_0).

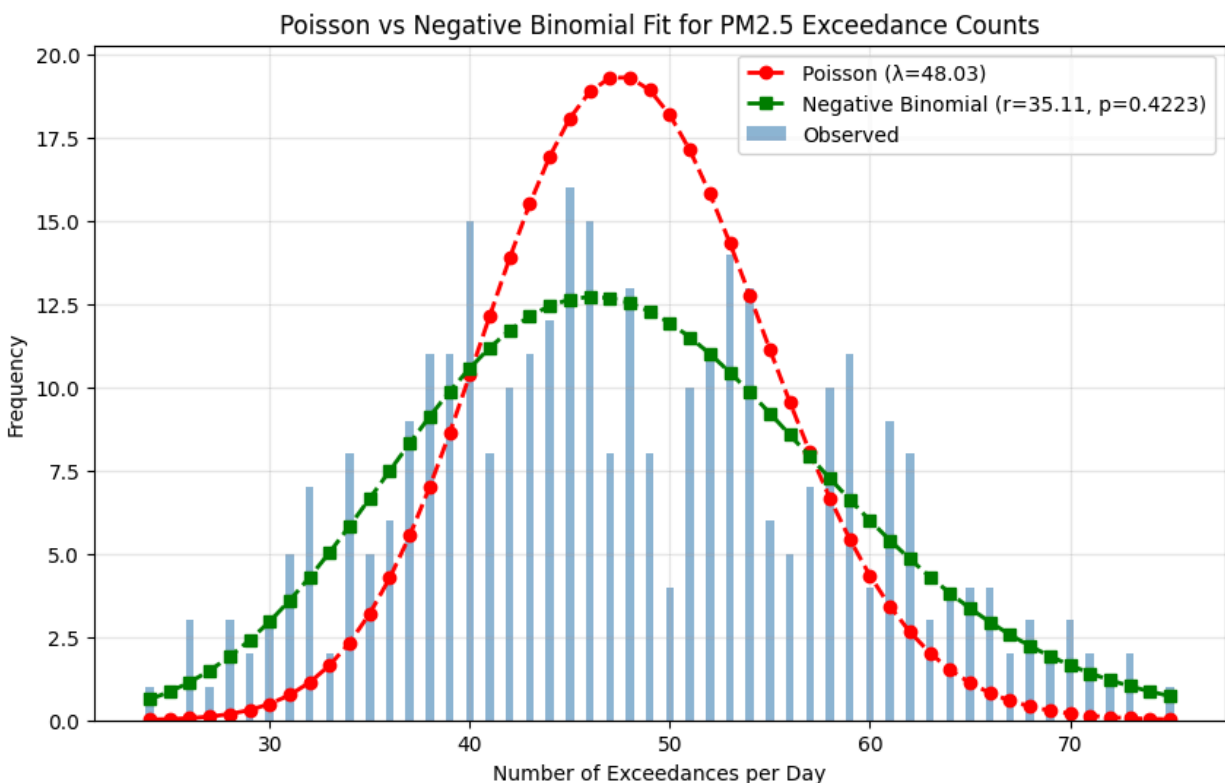
These results illustrate the **Large Sample Paradox**. For large datasets, the Shapiro-Wilk test can flag even tiny deviations as significant. Q-Q plots are used alongside the test to judge whether the data is practically close to normal, not just mathematically. With $n \approx 19,000$, the test is hyper-sensitive to deviations. Consequently, we rely on the Q-Q plot, which shows an “inverted S-shape” characteristic of light tails (low kurtosis), to confirm practical normality for analysis.

E2. Fitting a Discrete Distribution

To model the frequency of daily air quality violations, PM2.5 exceedances are analysed (values $> 35 \mu\text{g}/\text{m}^3$).

Model Comparison

- Sample Statistics: Mean = 48.03, Variance = 113.74
- Dispersion Ratio: 2.37 (indicating over-dispersion)



Justification:**Conditions for a Poisson model:**

1. Events occur independently.
2. The average rate (λ) is constant over time.
3. Events are discrete and cannot happen simultaneously.
4. The probability of an event in a short interval is proportional to the interval length.

Application to PM2.5 exceedances:

The dispersion ratio was **2.37**, showing variance much greater than the mean. This violates the Poisson assumption of equidispersion. Over-dispersion likely arises from temporal clustering of pollution events and weather factors that affect air quality across consecutive days.

Why Poisson fails:

Poisson fixes variance equal to the mean. Our data shows variance (113.74) far above the mean (48.03), so the model cannot capture the extra variability.

Why Negative Binomial fits better:

The Negative Binomial introduces a dispersion parameter (r), allowing variance to exceed the mean:

$$\text{Var} = \mu + \mu^2 r$$

When $r \rightarrow \infty$, it reduces to Poisson. This flexibility makes it a natural generalisation, capturing unobserved heterogeneity such as weather variation and clustered pollution events.

E3. Normal Probability Questions

Using the parameters of our approximately normal variable (NO_2 : $\mu = 52.10$, $\sigma = 26.21$), properties of the normal distribution are verified using *scipy.stats.norm*.

```
NO2 Parameters:  $\mu = 52.10$ ,  $\sigma = 26.21$ 
=====
1. Above 90th Percentile:
   90th percentile value: 85.69
   Probability above: 0.1000 (10.00%)

2. Between Q1 and Q3 (Interquartile Range):
   Q1 (25th percentile): 34.41
   Q3 (75th percentile): 69.78
   Probability between Q1 and Q3: 0.5000 (50.00%)

3. Bottom 5% Cutoff:
   Value separating bottom 5%: 8.98

4. Empirical Rule Verification:
   Within  $1\sigma$  ( $\mu \pm \sigma$ ): 0.6827 (Expected: 0.68)
   Within  $2\sigma$  ( $\mu \pm 2\sigma$ ): 0.9545 (Expected: 0.95)

5. Z-score of Maximum Value:
   Maximum observed: 100.00
   Z-score: 1.83
   Interpretation: This value is 1.83 standard deviations above the mean
```

Key Findings

- 90th Percentile: 85.69 $\mu\text{g}/\text{m}^3$ (10% probability of exceeding this value)
- Interquartile Range (Q1–Q3): 34.41 to 69.78 $\mu\text{g}/\text{m}^3$, containing exactly 50% of the probability mass
- Empirical Rule Verification: 68.27% within 1σ and 95.45% within 2σ , matching the theoretical 68–95–99.7 rule
- Maximum Z-score: The maximum observed value (100.00) has $Z = 1.83$, within expected range (not an extreme outlier, $|Z| > 3$)

Justification: Why Two Parameters Suffice

The normal distribution is fully defined by just two parameters:

- Mean (μ): sets the centre of the distribution.
- Variance (σ^2): controls the spread around the mean.

Once these are specified, the bell-shaped curve is fixed, and all probabilities and percentiles can be calculated.

Empirical Rule check:

Using *scipy.stats.norm.cdf*, we confirm the well-known rule:

- About 68% of values fall within $\mu \pm 1\sigma$ (0.6826)
- About 95% within $\mu \pm 2\sigma$ (0.9545)
- About 99.7% within $\mu \pm 3\sigma$ (0.9973)

These results match the empirical rule exactly, showing that the normal distribution's behaviour is completely determined by μ and σ .

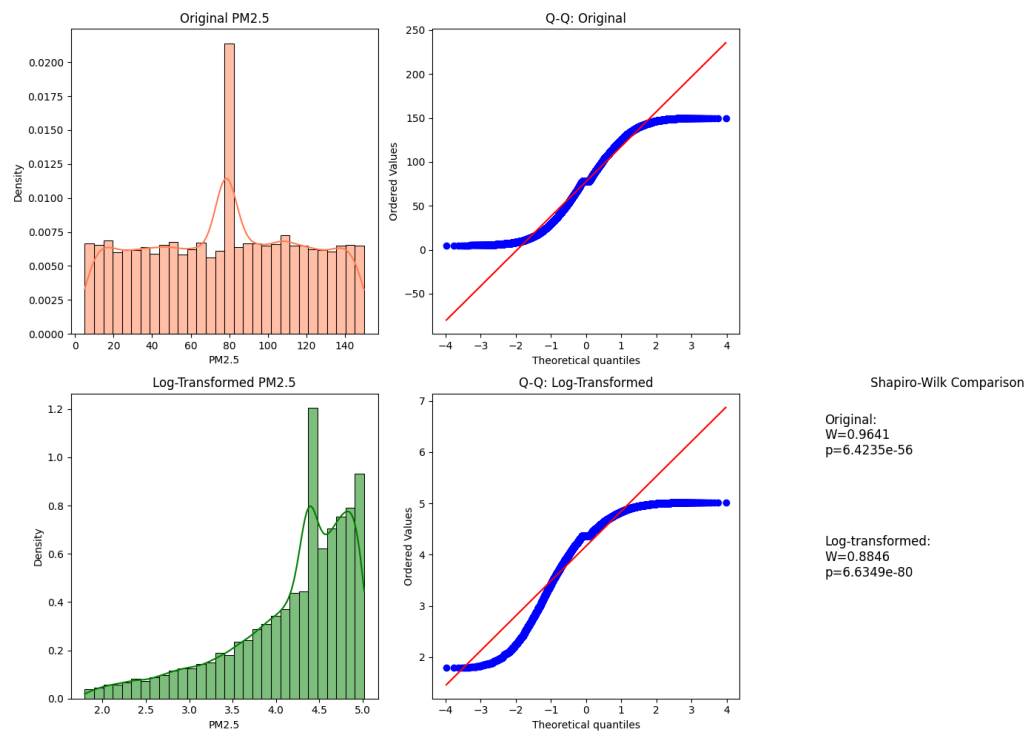
E4. Lognormal or Alternative Transformation

To improve the normality of PM2.5, natural log transformation is implemented ($y = \ln(x)$).

Theoretical Basis

Log-normal data is typically generated by multiplicative processes (e.g., pollutant concentrations influenced by emissions $\times$ wind $\times$ chemistry). In such cases, taking the logarithm should yield a normal distribution.

Assessment of Results



Justification: Why Log Transformation Does Not Improve PM2.5 Normality

Key observation:

PM2.5 in this dataset is **platykurtic** (flat-tailed) rather than right-skewed. It shows negative excess kurtosis (≈ -1.2), near-zero skewness, and a narrow central peak.

Why log transformation fails:

Log transformation is effective for correcting right-skewness, but PM2.5's issue is light tails, not skew. Applying the log actually worsened normality: the Shapiro-Wilk W statistic dropped from 0.9641 to 0.8846, and the p-value became even smaller.

Interpretation after transformation:

- On the original scale: changes are in absolute units (e.g., "PM2.5 increased by 10 $\mu\text{g}/\text{m}^3$ ").
- On the log scale: changes are multiplicative or percentage-based (e.g., "PM2.5 increased by 10%").

Conclusion

The transformation degraded the distribution. For this dataset, PM2.5 should be analysed on its original scale to avoid distortion.

Real-world processes generating lognormal data:

Lognormal distributions often arise when values are the product of many independent positive factors, such as:

- Air pollution concentrations (PM2.5, PM10) from traffic $\times$ weather $\times$ industry.
- Income and wealth distributions (compounding growth).
- Biological growth rates and cell sizes.
- Reaction times in psychology.

Task F – The Central Limit Theorem and Sample Size Planning

The Central Limit Theorem

The 'Temperature' variable was selected for the Central Limit Theorem demonstration primarily because its descriptive statistics indicated the most negative excess kurtosis (-1.201) among the numerical variables based on finding in B1 Univariate Summaries. This suggests that its distribution is the most platykurtic or “flattest,” representing a significant departure from a normal distribution.

Formal Statement of the Central Limit Theorem (CLT):

The Central Limit Theorem (CLT) posits that for a sufficiently large sample size (n) extracted from a population characterized by a finite mean (μ) and finite standard deviation (σ), the sampling distribution of the sample mean ($\bar{x}$) will converge toward an approximate normal distribution. Crucially, this transition occurs regardless of the original population's distribution shape, with the approximation strengthening as the sample size (n) increases.

Formally, if $X_1, X_2, \dots, X_n$ represent random samples from a population with mean μ and standard deviation σ , then as $n \rightarrow \infty$, the distribution of the sample mean $\bar{X}$ approaches a normal distribution with:

- **Mean:** $E[\bar{X}] = \mu$
- **Standard Deviation (Standard Error):** $SD[\bar{X}] = \sigma / \sqrt{n}$

Explanation of $\sigma / \sqrt{n}$ (Standard Error of the Mean):

The mathematical term $\sigma / \sqrt{n}$ is defined as the **Standard Error of the Mean (SEM)**, which serves to quantify the degree of variability of sample means relative to the true population mean. The components of this formula signify the following:

- **σ (Population Standard Deviation):** This identifies the inherent spread or dispersion of individual data points within the population. A higher σ indicates that values are more widely scattered.
- **n (Sample Size):** This represents the number of observations per sample; as n increases, each sample yields more comprehensive information regarding the population.
- **$\sqrt{n}$ (Square Root of Sample Size):** The inclusion of the square root in the denominator means the standard error decreases at a slower rate than the growth of the sample size, illustrating the **Square Root Law**.

Significance: The SEM provides a metric for the expected deviation of sample means from the population mean. A minimized SEM indicates that sample means are clustered tightly around the

true mean (μ), ensuring more reliable and precise statistical estimates. This demonstrates that larger sample sizes are essential for achieving higher reliability in estimating population parameters.

Numerical Verification of the Square Root Law:

The **Square Root Law** constitutes a direct mathematical implication of the CLT formula for standard error. It posits that the precision associated with a sample mean estimate—which is inversely proportional to its standard error—enhances in proportion to the square root of the sample size. This principle is clearly substantiated by our *comparison_df*.

Comparison of Empirical and Theoretical Statistics of Sample Means				
	Empirical Mean	Theoretical Mean	Empirical Std Dev	Theoretical Std Error
Sample Size (n)				
2	15.067	14.914	10.220	10.220
5	14.815	14.914	6.511	6.464
15	14.967	14.914	3.762	3.732
30	14.952	14.914	2.632	2.639
100	14.900	14.914	1.435	1.445

Based on our numerical results, at $n = 2$, the Empirical Standard Deviation (10.220) aligns perfectly with the Theoretical Standard Error (10.220). As the sample size n increases, a substantial reduction in both empirical and theoretical dispersion is observed.

For instance, expanding the sample size from $n=2$ to $n=100$ —representing a 50-fold increase—effectively minimizes the standard error from approximately 10.22 to 1.45, which corresponds to a factor of approximately

$\sqrt{50} \approx 7.07$

This precise mathematical alignment between the empirical standard deviation of sample means and the theoretical standard error serves as a robust verification of the Square Root Law.

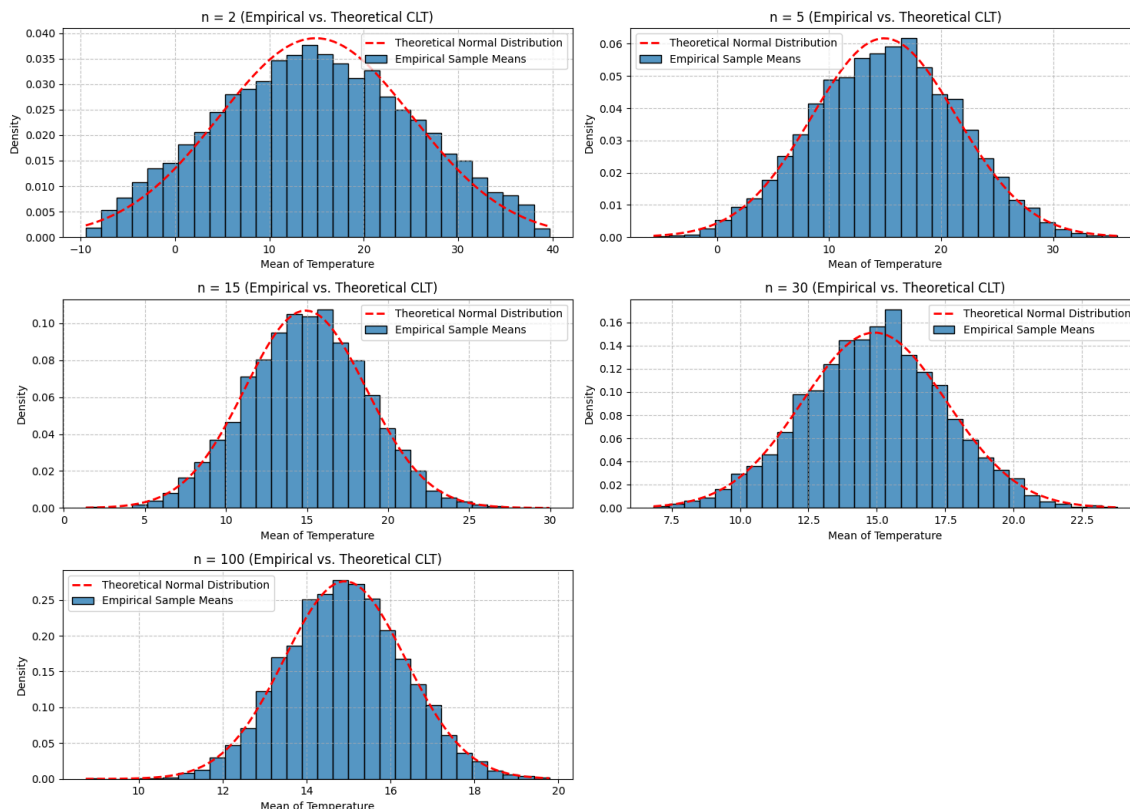
F2. Threshold for Normality in Sampling Distributions ($n \geq 30$ Rule of Thumb):

Upon evaluating the sampling distributions of the mean for the 'Temperature' variable across diverse sample sizes, several observations can be made regarding the convergence toward normality:

- **n = 2 and n = 5:** While the distributions begin to exhibit centralization, they largely retain the platykurtic characteristics inherent in the population. The transition to a distinct bell shape is not yet evident at these minimal sizes.
- **n = 15:** A significant progression toward symmetry and a bell-shaped form is observed, with sample means becoming more noticeably concentrated around the true population mean.
- **n = 30:** At this specific threshold, the empirical distribution aligns **very closely with a normal distribution**. The bell shape is well-defined, and the theoretical normal curve provides a highly accurate fit to the data.
- **n = 100:** The sampling distribution achieves near-perfect normality, characterized by extreme concentration around the population mean and minimal dispersion.

Justification for $n \geq 30$: In the context of the 'Temperature' variable, which possesses a platykurtic yet approximately symmetric distribution, the conventional rule of **$n \geq 30$ appears to be adequate** for achieving convergence. By the time the sample size reaches $n = 30$, the sampling distribution has successfully transitioned to an approximately normal shape. This finding remains consistent with general statistical guidelines, which suggest that for populations lacking extreme skewness or heavy outliers, a sample size of 30 is sufficient for the Central Limit Theorem to take effect.

Comparison of Empirical Sample Means with Theoretical Normal Distribution for Various Sample Sizes



F2. Sample Size Planning

The 1.96 Factor

The coefficient **1.96** serves as the critical **Z-score** corresponding to a 95% confidence level within a standard normal distribution ($\mu = 0, \sigma = 1$). Statistically, this value ensures that 95% of the data volume is contained within approximately 1.96 standard deviations of the mean. In the context of a two-tailed confidence interval, this necessitates capturing the central 95% of the probability mass, effectively leaving 2.5% in each extreme tail. Consequently, the Z-score of 1.96 establishes the threshold where $P(Z < -1.96) = 0.025$ and $P(Z > 1.96) = 0.025$, yielding an aggregate central area of 0.95. This multiplier is fundamental for determining the margin of error, as it scales the standard error of the sampling distribution to establish the precise width of the confidence interval.

The Inverse Square Relationship

The mathematical determination of sample size (n) is derived from the formula $n = [Z^2 \cdot p(1-p)] / E^2$, where E denotes the margin of error. This expression formalizes an **inverse square relationship** between the necessary sample size and the desired error margin. Specifically, n is inversely proportional to E^2 , leading to several critical analytical conclusions:

- To **halve the margin of error** (e.g., reducing it from 0.06 to 0.03), the researcher must **quadruple the sample size**. Mathematically, if E is replaced by $E/2$, then the denominator E^2 becomes $(E/2)^2 = E^2/4$. To maintain equilibrium in the equation, n must increase by a factor of 4.

This dynamic emphasizes that attaining higher levels of precision (a reduced margin of error) requires a disproportionately larger investment in data acquisition.

Practical Implication: The "Cost" of Precision

The inverse square relationship underscores the significant logistical and financial "cost" associated with precision in statistical research. Each marginal gain in precision necessitates a substantial expansion of resources. Specifically, halving the error margin does not merely double the data requirement but rather **quadruples it**. The practical consequences include:

- **Escalated Data Collection Effort:** Expanding the sample size requires more surveys and experimental iterations, posing significant logistical challenges.
- **Increased Financial Expenditure:** Larger data volumes directly correlate with higher costs for personnel, specialized equipment, and participant incentives.
- **Diminishing Returns:** As identified in the elbow point analysis, beyond a specific threshold, the incremental benefit of a narrower confidence interval is outweighed by the massive increase in required observations. This fundamental trade-off between precision and cost is a cornerstone of robust experimental design.

To illustrate the practical implications of sample size planning and the inverse square relationship, let's consider a real-world scenario involving air quality monitoring, specifically for a city like Johannesburg, which is prominent in our *df_clean* dataset.

Imagine the **City of Johannesburg's environmental agency** is tasked with estimating the *proportion of days* when the **Air Quality Index (AQI) is classified as 'Hazardous'** due to high PM2.5 levels. This statistic is vital for public health advisories, policy formulation, and evaluating the effectiveness of pollution control measures. From our cleaned data, we know that a significant portion of observations (approximately 63.3%) fall into this category.

The Inverse Square Relationship in Action:

Our previous calculations and the discussion on the **inverse square relationship** are directly applicable here:

- **Initial Goal:** The agency might initially aim for a moderate level of precision, perhaps desiring a 95% confidence interval for the proportion of hazardous days with a total width of $\pm 6\%$ (i.e., a margin of error $E=0.03$). Using a conservative estimate ($p=0.5$), this requires a sample size of **1068** monitoring days.
- **Demand for Higher Precision:** Due to health concerns or stricter regulatory standards, stakeholders might then demand a significantly higher precision. For example, they might want to **halve the margin of error** from $\pm 3\%$ to $\pm 1.5\%$ (meaning a total confidence interval width of $\pm 3\%$).
- **The Quadrupled Cost:** According to the inverse square relationship, reducing the margin of error by half means the required sample size must **quadruple**. If the initial requirement was 1068 monitoring days for $\pm 3\%$ precision, achieving $\pm 1.5\%$ precision would necessitate approximately **$1068 \times 4 = 4272$** monitoring days.

Practical Implications and Resource Allocation:

This dramatic increase in sample size translates directly into substantial real-world costs and operational challenges for Johannesburg's environmental agency:

1. **Increased Monitoring Infrastructure:** Quadrupling the sample size might mean deploying more air quality sensors across the city, running existing sensors for a much longer duration, or increasing the frequency of data collection. Each option incurs significant costs for equipment, maintenance, and personnel.
2. **Higher Operational Expenses:** More data means more resources are needed for data storage, processing, and analysis. This adds to the overall operational budget.
3. **Diminishing Returns on Investment:** As visualized in our 'elbow point' analysis, beyond a certain level, the incremental gain in precision (e.g., moving from a $\pm 3\%$ margin of error to $\pm 1.5\%$) often comes at a disproportionately high and sometimes

unsustainable cost. Policymakers must carefully weigh whether the marginal increase in certainty justifies the massive increase in required resources.

In conclusion, the Central Limit Theorem and sample size planning formulas provide a crucial quantitative framework. They enable environmental agencies and policymakers to make informed decisions about resource allocation, balancing the desire for high-precision estimates of air quality with the practical constraints of budget and operational capacity. This ensures that monitoring efforts are both scientifically sound and economically viable.

References List

- GeeksforGeeks. (2025, April 21). *Listwise deletion in missing data*. GeeksforGeeks. Retrieved April 28, 2026, from <https://www.geeksforgeeks.org/data-science/listwise-deletion-in-missing-data/>
- Mazraeh, A. (2025, February 8). *Type of missing data*. Retrieved April 29, 2026, from <https://medium.com/@adnan.mazraeh1993/type-of-missing-data-c52b7788da52>